

UNDERWATER ROBOTICS FOR INDUSTRIAL INSPECTION AND MAINTENANCE
(Applications of ROVs, AUVs, and Intelligent Robotic Systems for Inspection,
Monitoring, and Robotic Intervention)

BY

Eng. OVARA MARK

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CHAPTER ONE

INTRODUCTION

1.1 Introduction to Underwater Robotics for Industrial Inspection and Maintenance

A substantial proportion of industrial infrastructure is located in, passes through, or operates within underwater environments. Such infrastructure includes offshore oil and gas installations, subsea pipelines, underwater cables, offshore renewable energy systems, dams, ports, ship hulls, bridges, and other marine structures. The inspection and maintenance of these assets are necessary for identifying structural degradation, corrosion, cracks, leaks, marine growth, mechanical damage, and other conditions that can affect operational reliability and safety.

Inspection of underwater infrastructure presents several challenges because the working environment is fundamentally different from terrestrial environments. Limited visibility, water pressure, currents, corrosion, marine growth, restricted accessibility, and the distance between underwater assets and surface facilities can make inspection operations complex. These conditions have encouraged the development of remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs), and other underwater robotic systems capable of carrying sensors and inspection equipment into environments that are difficult or hazardous for human divers.

Underwater robotics therefore provides a means of extending inspection, monitoring, and maintenance capabilities beyond the limitations of direct human intervention. Modern underwater robotic systems combine propulsion, sensing, navigation, communication, control, and data-processing technologies to perform tasks ranging from visual inspection to more advanced forms of autonomous monitoring and robotic intervention.

1.2 Underwater Industrial Infrastructure

The increasing deployment of offshore energy, telecommunications, transportation, and marine infrastructure consequently creates a continuing requirement for inspection and maintenance technologies capable of operating underwater. Underwater infrastructure encompasses a wide range of assets that require regular inspection and maintenance. These include:

1.2.1 Oil and gas infrastructure

- subsea pipelines
- risers
- subsea production systems
- manifolds

- wellheads
- offshore platforms
- subsea valves
- storage and loading systems

1.2.2 Offshore renewable-energy infrastructure

- offshore wind turbine foundations
- subsea cables
- mooring systems
- floating platforms
- Tidal energy systems

1.2.3 Marine transportation infrastructure

- ship hulls
- ports
- docks
- quay walls
- underwater structures

1.2.4 Civil infrastructure

- dams
- bridges
- water-intake structures
- underwater foundations

1.2.5 Telecommunications infrastructure

- submarine communication cables
- cable landing infrastructure

1.3 Underwater Inspection and Maintenance

Inspection involves obtaining information about the condition of an underwater asset, while maintenance involves actions intended to preserve, repair, or restore its operational condition. Inspection activities can include:

- visual examination

- corrosion detection
- crack detection
- thickness measurement
- leak detection
- dimensional measurement
- structural assessment
- marine-growth assessment
- subsea equipment monitoring

1.3.1 Maintenance activities can include:

- cleaning
- removal of marine growth
- valve operation
- manipulation
- repair
- cutting
- welding
- tooling
- replacement of components

The distinction between inspection and intervention is important for underwater robotics because the robotic requirements are different. A vehicle designed primarily to collect images does not necessarily possess the mechanical capability required to manipulate or repair infrastructure. Consequently, underwater robotic systems can be considered across a progression:

Observation → Inspection → Monitoring → Diagnosis → Intervention → Maintenance

This progression establishes the basis for examining ROVs, AUVs, and intelligent robotic systems.

1.4 Remotely Operated Vehicles (ROVs)

A remotely operated vehicle is an underwater robotic platform connected to a surface vessel or control station through a communication and power umbilical. The umbilical can provide:

- electrical power
- communication
- video transmission
- control signals
- data transmission

An operator located at the surface controls the vehicle while receiving information from onboard cameras and sensors. The ROVs can be equipped with:

- cameras
- lights
- sonar
- depth sensors
- inertial sensors
- manipulators
- hydraulic or electric tools
- cleaning equipment
- inspection sensors

This makes ROVs particularly useful for tasks where continuous operator supervision and physical interaction with infrastructure are required. They are currently used extensively in offshore inspection and intervention because they can operate for extended periods while carrying substantial payloads and tooling. Current ongoing research into subsea inspection continues to investigate improvements in ROV perception, control, navigation, and intervention capabilities.

1.5 ROV Applications in Industrial Inspection

1.5.1 Pipeline inspection

- corrosion
- mechanical damage
- leaks
- deformation
- marine growth
- structural anomalies

1.5.2 Platform inspection

- structural members
- underwater foundations
- risers
- subsea equipment
- weld regions

1.5.3 Ship inspection

- hulls
- propellers
- rudders
- underwater fittings

1.5.4 Subsea equipment

ROVs can inspect and interact with:

- valves
- manifolds
- wellheads
- connectors
- control systems

This ability to combine sensing and manipulation is one of the major distinctions between inspection oriented ROVs and purely observational underwater vehicles.

1.6 Autonomous Underwater Vehicles (AUVs)

Unlike ROVs, AUVs operate without a physical tether connecting them to a surface control station. An AUV generally contains:

- onboard batteries
- propulsion systems
- navigation sensors
- computers
- cameras
- sonar
- inertial measurement units
- acoustic positioning systems
- mission-planning software

The vehicle is programmed with a mission and performs navigation and data collection autonomously or semi-autonomously. This provides an important difference:

ROV: Human operator → tether → underwater vehicle; AUV: Mission plan → onboard autonomy → underwater vehicle.

AUVs are therefore particularly suitable for large area inspection and data collection where continuous tethered operation is unnecessary.

1.8 AUV Applications in Industrial Inspection

- pipeline surveys
- seabed mapping
- offshore infrastructure inspection
- underwater archaeological surveys
- environmental monitoring
- cable-route surveys
- structural mapping

1.9 Sensors for Underwater Robotic Inspection

The effectiveness of an underwater inspection robot depends heavily on its sensing system. Because underwater visibility can be poor, no single sensor is sufficient for every inspection task. Important sensor categories include:

1.9.1 Optical cameras

- visual inspection
- object identification
- marine-growth assessment
- surface-condition analysis

1.9.2 Sonar

Sonar systems transmit and receive acoustic signals to detect objects and structures underwater. They are particularly important when:

- visibility is poor
- the robot operates at depth
- long-range detection is required

Types include:

- multibeam sonar
- imaging sonar
- Side scan sonar
- Forward looking sonar

1.9.3 Inertial Measurement Units

They contribute to vehicle navigation and motion estimation, while it measure:

- acceleration
- angular velocity
- orientation

1.9.4 Depth/pressure sensors

These provide measurements of vehicle depth and pressure.

1.9.5 Acoustic positioning

Systems such as USBL, LBL, and related acoustic positioning technologies can provide underwater localization where GPS is unavailable.

1.9.6 Environmental sensors

Depending on the mission, robots can carry sensors for:

- temperature
- salinity
- dissolved oxygen
- turbidity
- chemical detection

The combination of multiple sensors creates a multimodal perception system. These technologies are used currently in most underwater robotic systems.

1.10 Navigation and Localization

Navigation is one of the fundamental challenges in underwater robotics. Unlike terrestrial and aerial robots, underwater robots cannot normally rely on conventional GPS while submerged. Consequently, underwater robots use combinations of:

- inertial navigation
- Doppler velocity logs
- acoustic positioning
- depth sensors
- sonar
- visual odometry
- simultaneous localization and mapping (SLAM)

Underwater SLAM research combines sensing and estimation techniques to enable robots to construct representations of their environment while simultaneously estimating their position.

The general process can be represented as: **Sensor measurements → Localization → Mapping → Environment representation → Path planning → Robot control**. This capability is particularly important for autonomous inspection because an AUV must know both **where it is** and **where the inspected infrastructure is located**.

1.11 Computer Vision and Artificial Intelligence

The increasing availability of high-resolution cameras, sonar, onboard computing, and machine-learning methods has shifted underwater inspection from simple data collection toward automated interpretation. Traditional inspection can require an operator to examine large quantities of video or sonar data manually. Artificial intelligence can instead be used for:

- object detection
- corrosion identification
- crack detection
- anomaly detection
- marine-growth classification
- pipeline recognition
- structural segmentation
- defect classification

Deep learning based computer vision has increasingly been investigated for underwater inspection because conventional visual processing can be affected by turbidity, lighting variation, color distortion, and image degradation. The resulting architecture can be expressed as: **Camera/Sonar → Data acquisition → AI model → Defect detection → Condition assessment**. This creates a transition from **robotic data collection** to **robotic perception and decision making**.

1.12 Intelligent Underwater Robotic Systems

The development of intelligent underwater systems involves integrating sensing, navigation, perception, planning, and control. An intelligent inspection system can therefore operate through the following sequence: **Perception → Localization → Object/defect identification → Mission planning → Motion planning → Inspection → Decision → Intervention**

This represents an important transition from remotely controlled vehicles toward autonomous or semi-autonomous robotic systems. Research into autonomous underwater inspection increasingly combines machine learning, computer vision, sensor fusion, autonomous navigation, and robotic manipulation.

1.13 Robotic Intervention and Maintenance

Inspection alone does not restore an underwater asset. Consequently, research is increasingly investigating underwater robotic intervention. Intervention can include:

- cleaning
- grasping
- valve operation
- tool positioning
- cutting
- grinding
- component manipulation
- repair operations

ROVs with robotic manipulators are already used for intervention tasks in subsea operations. The development of autonomous manipulation introduces additional requirements because the robot must understand the location and geometry of the target before interacting with it. A typical intervention system consists of:

- ROV
- Manipulator
- Force/position sensing
- Camera/sonar
- Tool
- Human or autonomous controller

1.14 From Inspection to Predictive Maintenance

The combination of robotic inspection, sensors, data processing, and artificial intelligence creates the possibility of moving from periodic inspection toward condition-based and predictive maintenance. The traditional approach can be represented as: **Scheduled inspection → defect discovered → maintenance performed**

An intelligent approach can instead use: **Continuous/periodic monitoring → data analysis → condition assessment → degradation prediction → maintenance planning**. This requires the robotic system to collect consistent data over time e.g: **Inspection 1** → corrosion measurement, **Inspection 2** → corrosion measurement, **Inspection 3** → corrosion measurement → **Trend analysis → Degradation estimation → Maintenance decision**. This creates a direct connection between underwater robotics and industrial asset management.

1.15 Current Trends in Underwater Robotic Inspection

Current underwater robotics research is moving toward systems that combine **autonomy, artificial intelligence, advanced sensing, improved navigation, and robotic intervention**.

One important trend is the development of autonomous inspection systems capable of performing missions with reduced dependence on continuous human control.

Another is the integration of multiple sensors. Cameras, sonar, inertial systems, acoustic positioning, and other sensors can be combined to provide more reliable environmental perception. A further trend is the use of artificial intelligence for automated defect detection and interpretation of inspection data.

Research is also moving toward **resident subsea robotics**, where robotic systems remain close to underwater infrastructure for extended periods rather than being deployed only during individual inspection campaigns. This changes the concept of underwater inspection from an occasional activity to a more continuous monitoring process. Resident systems can potentially support:

- continuous monitoring
- repeated inspection
- anomaly detection
- rapid response
- reduced deployment frequency

1.16 Future Trends in Underwater Robotics

Future underwater inspection systems are expected to increasingly combine autonomous navigation, artificial intelligence, multi-robot systems, advanced sensing, and intervention capabilities.

1.16.1 Increased autonomy

Future AUVs are expected to perform increasingly complex missions with less direct human supervision. The progression can be represented as: **Remote control → assisted autonomy → supervised autonomy → autonomous operation**

1.16.2 AI-based inspection

AI can increasingly support automated:

- defect detection
- image interpretation
- anomaly classification

- infrastructure recognition
- mission planning

1.16.3 Multi-robot systems

Multiple underwater robots can potentially divide inspection tasks across a large infrastructure network, this can reduce the dependence on a single vehicle. E.g :

Robot A → Pipeline section 1

Robot B → Pipeline section 2

Robot C → Platform structure

↓

Shared inspection database

1.16.4 AUV and ROV cooperation

AUVs can conduct broad autonomous surveys while ROVs can perform detailed inspection and physical intervention. This can represent a progression from **detection to action**. The resulting architecture can be:

AUV → large-area survey

↓

AI → anomaly detection

↓

ROV → detailed inspection

↓

Manipulator → intervention

1.17 Energy and Endurance in Underwater Robots

Energy remains a fundamental limitation for autonomous underwater systems. AUVs generally rely on onboard energy storage, which limits:

- mission duration

- propulsion time
- sensor operation
- computing
- Communication

Energy consumption is distributed among: **Propulsion + navigation + sensors + computing + communication**. Consequently, improvements in battery technology alone do not address every aspect of underwater robotic endurance. Current research is investigating alternative energy approaches including:

- improved batteries
- fuel cells
- energy-efficient propulsion
- underwater docking
- wireless charging
- wave energy
- ocean-current harvesting
- thermal-gradient energy harvesting
- solar energy for suitable shallow water/surface applications

1.18 Challenges in Underwater Robotic Inspection

1.18.1 Limited visibility

Turbidity, suspended particles, lighting conditions, and marine growth can reduce optical inspection quality.

1.18.2 Underwater localization

GPS cannot normally provide direct positioning while the robot is submerged, requiring alternative navigation technologies.

1.18.3 Communication

Radio communication is severely limited underwater. Acoustic communication is therefore commonly used but has limitations in bandwidth, latency, and reliability.

1.18.4 Hydrodynamic disturbances

Currents can affect:

- vehicle stability

- trajectory
- energy consumption
- inspection accuracy
- manipulator positioning

1.18.5 Corrosion and biofouling

Marine environments can degrade robotic components and sensors.

1.18.6 Energy limitations

Energy spent across different sections lead to limitations because AUVs must balance: **mission duration ↔ propulsion energy ↔ sensing ↔ computation ↔ communication.**

1.18.7 Autonomous manipulation

Physical intervention requires the robot to understand the target's position, orientation, geometry, and interaction forces.

1.18.8 AI reliability

Machine learning systems can be affected by:

- limited underwater datasets
- environmental variation
- poor image quality
- domain differences
- false detections

1.19 Research Objectives

- To investigate the applications of ROVs and AUVs in underwater industrial inspection and monitoring.
- To examine the sensing and navigation technologies used by underwater robots for detecting, locating, and assessing defects in subsea infrastructure.
- To investigate the application of artificial intelligence and computer vision for automated underwater inspection and anomaly detection.
- To examine robotic manipulation and intervention technologies applicable to underwater maintenance operations.
- To investigate the integration of autonomous navigation, perception, inspection, and intervention into intelligent underwater robotic systems.

- To evaluate the energy, communication, environmental, and operational constraints affecting autonomous underwater inspection systems.
- To develop or simulate a conceptual underwater robotic inspection system capable of performing selected inspection or monitoring tasks.

1.20 Research Questions

- What are the current applications of ROVs and AUVs in industrial underwater inspection and monitoring?
- What sensing and navigation technologies enable underwater robots to detect, locate, and assess defects in subsea infrastructure?
- How can artificial intelligence and computer vision improve the detection and classification of underwater structural defects?
- What robotic manipulation and intervention technologies can be applied to underwater maintenance operations?
- How can perception, navigation, inspection, and intervention capabilities be integrated into an intelligent underwater robotic system?
- What energy, communication, environmental, and operational factors limit the autonomy and endurance of underwater inspection robots?

1.21 Significance of the Research

Underwater infrastructure represents a significant component of modern energy, transportation, telecommunications, and marine industries. The ability to inspect and maintain such infrastructure efficiently is therefore important for maintaining operational reliability.

Underwater robotics provides an approach for conducting inspection and maintenance while reducing dependence on direct human presence in difficult underwater environments.

The integration of autonomous navigation, advanced sensing, artificial intelligence, and robotic intervention could further transform underwater robotics from remotely controlled inspection platforms into intelligent systems capable of collecting, interpreting, and acting upon information about underwater infrastructure.

The research therefore provides a framework for examining the transition from underwater observation to intelligent inspection, monitoring, diagnosis, and robotic intervention.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Underwater robotic systems have developed from remotely operated platforms primarily used as extensions of human divers into increasingly autonomous systems capable of sensing, navigation, inspection, monitoring, manipulation, and intervention. This development has been driven largely by the difficulty of performing underwater operations using conventional human based approaches. Subsea infrastructure such as pipelines, offshore platforms, subsea cables, marine renewable energy installations, dams, ship hulls and other structures must be inspected periodically for corrosion, cracks, deformation, marine growth, leakage and mechanical damage. Recent systematic reviews identify unmanned underwater vehicles (UUVs), including remotely operated vehicles (ROVs) and autonomous underwater vehicles (AUVs), as increasingly important platforms for structural inspection and underwater condition assessment. ([ScienceDirect](#))

The literature, however, shows that underwater inspection is not simply a problem of placing a camera on an underwater vehicle. A useful inspection system must first determine where the vehicle is, understand its surroundings, maintain an appropriate position relative to the asset, acquire usable inspection data, identify relevant anomalies, and ultimately convert those observations into information that can support maintenance decisions. These requirements create an interconnected problem involving vehicle dynamics, sensing, localization, navigation, perception, artificial intelligence, communication, energy management and, for intervention systems, robotic manipulation.

This chapter therefore reviews the development of ROV and AUV based inspection systems, the sensing and navigation technologies used for subsea inspection, the application of computer vision and artificial intelligence to underwater anomaly detection, robotic manipulation and intervention, and the integration of these technologies into increasingly autonomous underwater robotic systems.

2.2 ROVs and AUVs in Underwater Industrial Inspection and Monitoring

The distinction between ROVs and AUVs is fundamental to understanding underwater robotic inspection. ROVs are generally connected to a surface vessel through an umbilical cable that provides power, communication and control. This connection allows continuous transmission of high bandwidth sensor data and enables an operator to directly supervise the vehicle. Work class ROVs can also carry relatively large manipulators and specialised tooling, making them particularly suitable for complex subsea intervention tasks.

AUVs, in contrast, operate without a physical connection to the surface and therefore carry their own power, sensing, computation and navigation systems. This gives them greater freedom of movement and removes the physical limitations imposed by an umbilical. However, autonomy also introduces significant challenges involving energy capacity, underwater localization, communication, decision making and fault handling.

This distinction has influenced how the two platforms have traditionally been used. ROVs have been widely adopted for inspection and intervention because an operator can observe the environment and control the vehicle and manipulator in real time. AUVs have historically been more strongly associated with surveying, mapping, bathymetry and data collection. The development of intervention AUVs represents an attempt to extend autonomous vehicles from observation toward physical interaction with subsea infrastructure. Ridao et al. describe this transition as a move toward vehicles capable not only of surveying but also of tasks such as valve turning, connector manipulation and object recovery. ([ScienceDirect](#))

Recent research increasingly considers these platforms as part of a wider spectrum rather than completely separate technologies. Lightweight inspection ROVs can provide relatively low cost visual inspection, while work class ROVs provide greater payload and manipulation capability. AUVs can undertake longer autonomous surveys without an umbilical, while intervention AUVs seek to combine autonomous mobility with manipulation. A 2025 review of AUV technology identifies power, planning and control, perception, navigation, communications and artificial intelligence as interconnected technologies determining the ability of AUVs to undertake increasingly complex missions. ([ScienceDirect](#))

The industrial importance of these capabilities is particularly evident in offshore oil and gas and offshore renewable energy. Robots have been investigated for subsea inspection, monitoring, maintenance and asset integrity, with subsea residency and autonomous operation being explored as ways of reducing dependence on conventional vessel based operations. ([ScienceDirect](#))

A recent systematic review of UUV based structural health monitoring similarly categorises the literature into two broad areas: improvements to underwater vehicle capabilities and improvements to instrumentation for structural health monitoring. This distinction is important because an inspection robot is only useful when the vehicle and its sensing system operate effectively together. ([ScienceDirect](#))

2.3 Sensing Technologies for Underwater Inspection

The effectiveness of an underwater inspection robot depends strongly on its ability to perceive both its environment and the asset being inspected. Unlike terrestrial robots, underwater systems operate in an environment where electromagnetic signals are severely attenuated, optical

visibility can deteriorate rapidly, and environmental conditions can change considerably with depth, turbidity, lighting and water movement.

Consequently, underwater inspection systems commonly combine several sensor modalities rather than relying on a single sensor.

2.3.1 Optical Cameras

Optical cameras remain one of the most important sensing technologies for visual underwater inspection because they can provide detailed information about surface condition. Cameras can potentially reveal cracks, corrosion, marine growth, coating damage, deformation and other visible abnormalities.

However, underwater imaging introduces several problems. Light absorption and scattering can produce colour distortion, low contrast, haze and loss of fine details. These effects become particularly severe in turbid water or low light environments. Underwater image enhancement research therefore focuses on restoring contrast, colour and structural details before images are passed to subsequent computer-vision algorithms. Anwar and Li's survey of underwater image enhancement demonstrates that deep-learning approaches have become an important research direction, while also identifying limitations associated with datasets, evaluation methods and generalisation. ([ScienceDirect](#))

This creates an important relationship between sensing and artificial intelligence, poor quality image acquisition can directly affect the reliability of automated defect detection.

2.3.2 Sonar

Sonar provides an alternative or complementary sensing modality when optical cameras cannot provide sufficient visibility. Acoustic sensing can operate over greater distances and is less dependent on ambient light and water clarity.

This is particularly relevant to infrastructure inspection because sonar can assist with detecting and locating structures before the vehicle approaches them. Imaging sonar can also support object detection and tracking in environments where optical visibility is poor.

However, sonar data generally has lower spatial detail than high quality optical imagery and can be affected by acoustic noise, multipath propagation and environmental conditions. Consequently, many modern systems investigate **sensor fusion**, combining sonar, optical cameras, inertial measurements and other sensors rather than attempting to make one sensor perform every task.

A 2026 review of subsea pipeline and cable inspection specifically identifies vision, sonar, electromagnetic/magnetic sensing and sensor fusion as major approaches for tracking and inspecting underwater linear infrastructure. ([ScienceDirect](#))

2.3.3 Inertial and Velocity Sensors

Inertial Measurement Units (IMUs), Doppler Velocity Logs (DVLs) and pressure sensors are important for determining the vehicle's motion and position.

An IMU measures acceleration and angular velocity, allowing the vehicle's orientation and movement to be estimated. However, inertial navigation is subject to accumulated error over time.

DVL systems can estimate vehicle velocity relative to the seabed or nearby surfaces using Doppler measurements. Pressure sensors provide depth information. These measurements can be combined through sensor-fusion algorithms to produce more stable estimates of the vehicle's state.

The 2023 review of AUV navigation identifies integrated navigation as an important approach because no individual navigation sensor can completely solve underwater localization under all conditions. ([ScienceDirect](#))

2.4 Navigation and Localization for Underwater Inspection

Navigation is one of the major differences between underwater robotics and terrestrial robotics. An autonomous vehicle operating outdoors on land can normally use GPS for global positioning. An underwater vehicle cannot continuously depend on GPS because electromagnetic signals do not propagate effectively through water.

Consequently, underwater vehicles must construct their position estimates from alternative technologies. Traditional approaches include:

- Dead reckoning
- Inertial navigation systems
- DVL
- USBL
- SBL
- LBL
- Pressure/depth sensing
- Acoustic positioning
- Visual odometry
- Sonar-based localization

- SLAM
- Multi sensor fusion

Acoustic positioning systems such as Long Baseline (LBL), Short Baseline (SBL) and Ultra Short Baseline (USBL) can provide external positioning information. However, they introduce additional infrastructure and can be affected by sound-speed variations, multipath effects and environmental conditions. ([MDPI](#))

2.4.1 Underwater SLAM

Simultaneous Localization and Mapping (SLAM) has become particularly important because it allows an underwater robot to estimate its position while constructing a representation of the environment.

Visual SLAM uses camera observations to estimate vehicle motion and construct maps. However, underwater visual SLAM is complicated by turbidity, illumination variation, suspended particles, low texture and image degradation.

Zhang et al. describe underwater visual SLAM as a challenging problem resulting from unstructured underwater environments and the absence of GPS. Their review covers visual odometry, state optimisation and loop-closure detection and shows the progression from traditional filtering based approaches toward optimisation-based approaches. ([ScienceDirect](#))

More recently, bathymetric SLAM has also received attention, particularly for AUV navigation in unstructured environments. Recent review work identifies front end odometry, loop closure detection and back-end optimisation as key components of bathymetric SLAM systems. ([ScienceDirect](#))

For infrastructure inspection, accurate localization is especially important because the robot must associate an observed defect with a physical location on the structure. Detecting a crack without being able to determine where that crack is located has limited value for maintenance planning.

2.5 Artificial Intelligence and Computer Vision for Automated Inspection

The move from human operated inspection toward autonomous inspection depends heavily on the ability of robots to interpret sensor data.

Traditional inspection involves a human operator viewing images or video and deciding whether an observed feature represents corrosion, cracking, deformation, marine growth or another anomaly. This approach can be effective but becomes difficult when large quantities of inspection data are generated.

Artificial intelligence provides the possibility of automating parts of this process. Computer-vision systems can broadly perform four related tasks:

- Image enhancement
- Object or defect detection
- Semantic/instance segmentation
- Quantitative condition assessment

Underwater image enhancement is particularly important because underwater images often contain colour distortion, haze, low contrast and blurred structures. Deep-learning-based enhancement methods have therefore been investigated to improve the quality of images before detection or segmentation. A 2024 survey categorises deep-learning approaches according to network architecture, learning strategy, training stage and other methodological factors, while highlighting the continuing difficulty of comparing models across different datasets and experimental conditions. ([arXiv](#))

Once an image has been enhanced, detection algorithms can be used to locate potential defects. Recent computer vision literature on pipeline inspection identifies cracks, corrosion, deformation and dents as major targets for automated detection. Deep learning methods can achieve strong detection performance under suitable conditions, but their generalisation remains strongly dependent on dataset quality, quantity and preprocessing. ([ScienceDirect](#))

This is a major issue for underwater robotics.

A model trained using clear images of one type of pipeline may not perform equally well on another pipeline covered in marine growth or recorded under different lighting and turbidity conditions.

2.5.1 Segmentation and Defect Quantification

Detection answers an important question: **Where is the suspected defect?** Segmentation goes further by identifying the pixels belonging to the defect or structure. This makes segmentation particularly valuable when the objective is not simply to identify a defect but to estimate its size, shape or extent.

Recent reviews of computer vision based pipe inspection distinguish between 2D defect detection and 3D reconstruction and segmentation. They also identify preprocessing, computer aided identification and post-processing as important stages within the inspection pipeline. Emerging approaches include Transformer-based and semi-supervised learning methods. ([ScienceDirect](#))

This progression is important for robotic inspection because condition assessment ultimately requires more than a binary classification such as *defect/no defect*. Maintenance

decisions may require information about defect dimensions, location, severity and change over time.

A recent systematic study of optical inspection of underwater critical infrastructure similarly considers the complete pipeline from robotic data acquisition and image enhancement through damage detection, segmentation, defect quantification and 3D reconstruction. ([ScienceDirect](#))

2.6 Robotic Manipulation and Underwater Intervention

Inspection represents only one level of underwater robotic capability. Once a robot can identify a problem, the next question is whether it can physically interact with the infrastructure to perform a task.

This has led to research into **Underwater Vehicle Manipulator Systems (UVMS)**, where a vehicle is equipped with one or more robotic manipulators. Underwater manipulators have been investigated for tasks including:

- Valve operation
- Connector plugging and unplugging
- Object recovery
- Sampling
- Cleaning
- Tool deployment
- Maintenance
- Repair
- Inspection with contact-based sensors

Sivčev et al. provide a comprehensive review of underwater manipulators and discuss electrically and hydraulically actuated systems, manipulator control, motion planning and the practical limitations affecting subsea manipulation. ([ScienceDirect](#))

The manipulator introduces another layer of complexity because the vehicle itself is not stationary. When a robotic arm moves, the resulting reaction forces can disturb the vehicle's position and orientation.

Therefore, underwater manipulation involves a coupled **vehicle manipulator control problem** rather than simply controlling a conventional industrial robotic arm.

The challenge becomes particularly significant when the vehicle is free floating. Contact with the infrastructure can generate forces that move the vehicle, while currents can simultaneously disturb both the vehicle and the manipulator.

2.7 From ROV Intervention to Intervention AUVs

Traditional ROV based intervention usually keeps a human operator in the control loop. This provides an important advantage: the operator can interpret unexpected situations and modify the manipulation strategy.

However, this approach also creates dependence on high bandwidth communication and skilled operators.

Marani et al. identify communication bandwidth and delay as major obstacles to remote underwater manipulation. Acoustic communication, in particular, does not provide the same bandwidth and responsiveness available to terrestrial systems. ([ScienceDirect](#))

This has motivated the development of **Intervention Autonomous Underwater Vehicles (I-AUVs)**.

Ridao et al. reviewed the development of I-AUVs and identified projects including ALIVE, SAUVIM and GIRONA 500. Demonstrated capabilities included autonomous valve turning, connector manipulation and object recovery. ([ScienceDirect](#))

More recent research continues to move toward autonomous inspection, maintenance and repair. The SEAVENTION project, for example, investigated autonomous subsea intervention for inspection, maintenance and repair applications and proposed a three level taxonomy of autonomy:

- Mission-level autonomy
- Task-level autonomy
- Vehicle-level autonomy

The project also investigated 6D pose estimation of subsea objects using 2D and 3D cameras, including marker-based and markerless approaches. ([ScienceDirect](#))

This is significant because successful intervention requires the robot to know not only **what** the target is but also its complete spatial pose.

2.8 Integration of Navigation, Perception, Inspection and Intervention

One of the strongest themes emerging from the literature is that underwater autonomy cannot be achieved by improving individual subsystems independently.

- A vehicle may have excellent navigation but poor perception.
- It may have excellent computer vision but insufficient localization accuracy.
- It may identify a valve accurately but lack the manipulation capability to operate it.

- It may perform all these tasks but run out of energy before completing the mission.
- Therefore, autonomous underwater inspection should be understood as an integrated system.

A simplified architecture can be represented as:

- **Mission Planning**
- **Navigation + Localization**
- **Perception + Sensor Fusion**
- **Infrastructure Detection**
- **Inspection / Anomaly Detection**
- **Decision Making**
- **Manipulation / Intervention**
- **Verification + Reporting**

Recent work on full autonomy in underwater robotic systems identifies navigation, perception, communication, energy management, control and decision-making as interconnected components. The literature indicates that many current systems still operate with partial rather than complete autonomy, while sensor fusion and AI-based decision-making are being investigated to increase adaptability. ([ScienceDirect](#))

The 2025 review of AUV technologies reaches a similar conclusion by treating power systems, planning and control, perception, navigation, communication and AI as key technologies that must advance together for more complex AUV applications. ([ScienceDirect](#))

2.9 Energy Constraints in Autonomous Underwater Inspection

Energy is one of the most important limitations affecting AUV operation. An autonomous underwater vehicle must supply energy to propulsion, sensors, onboard computing, navigation, communication, actuators and other subsystems.

Unlike ROV, which can receive power through its tether, an AUV must carry its energy source onboard. This creates a fundamental trade-off:

- **More battery capacity → greater endurance but increased mass.**
- **More propulsion → greater mobility but greater energy consumption.**
- **More computation → improved autonomy but increased electrical demand.**
- **More sensors → better perception but additional power and payload requirements.**

Recent reviews of AUV technology identify power systems as one of the major areas influencing operational capability. ([ScienceDirect](#))

Energy limitations also affect navigation and path planning. A recent review of AUV path planning notes that selecting the shortest geometric route is not necessarily equivalent to selecting the lowest-energy route. Vehicle motion against currents, for example, can substantially increase propulsion requirements. ([MDPI](#))

This creates an interesting connection between navigation and energy management. A truly autonomous inspection system could eventually select a route not only according to: **“Where do I need to go?”** but also: **“Which route allows me to complete the inspection using the available energy?”**

Research into underwater wireless power transfer also reflects the difficulty of extending AUV mission duration. Recent work reviews resonant inductive, magnetic-resonance and acoustic approaches for underwater power transfer, with alignment, conductivity and energy-loss challenges remaining important considerations. ([ScienceDirect](#))

2.10 Communication Constraints

Radio frequency communication, which is widely used by terrestrial robots, does not propagate efficiently through seawater. Underwater systems therefore commonly rely on acoustic communication when longer range communication is required. However, acoustic communication introduces limitations involving:

- Low bandwidth
- Limited data rate
- High latency
- Noise
- Multipath propagation
- Variable channel conditions

A review of collaborative AUV systems identifies communication, localization and navigation as major challenges because GPS and conventional radio communication cannot be directly relied upon underwater. Acoustic communication offers greater range than optical communication but at the cost of lower data rates and higher latency. ([MDPI](#))

This becomes particularly important for teleoperated ROVs and autonomous intervention. A human operator controlling a robotic arm requires timely feedback. If communication introduces significant delay, the operator's ability to respond to changing contact conditions is reduced.

This is one reason why increasing autonomy is attractive for intervention missions: the vehicle can potentially perform time sensitive local control without waiting for every command to travel between the vehicle and surface operator.

Recent reviews of underwater wireless communication continue to investigate acoustic, optical and emerging communication technologies, with energy efficiency and communication performance remaining important trade-offs for AUV applications. ([MDPI](#))

2.11 Environmental and Operational Constraints

- Turbidity reduces optical visibility and affects camera-based perception.
- Pressure increases with depth and places requirements on vehicle housings, connectors, sensors and actuators.
- Currents affect vehicle position, propulsion requirements and energy consumption.
- Biofouling can obscure infrastructure surfaces and reduce the effectiveness of visual inspection.
- Corrosion affects both the infrastructure being inspected and potentially the robotic system itself.
- Temperature can influence batteries, electronics, sensors and mechanical components.
- Communication constraints make continuous remote supervision difficult for autonomous systems.
- AUVs must complete their missions within finite onboard energy resources.

These factors interact rather than appearing independently. For example, poor visibility may force an AUV to move closer to a structure. Moving closer may increase collision risk. Maintaining position against a current may increase propulsion energy. Increased energy consumption can shorten mission duration. If communication is also limited, the vehicle may need greater onboard autonomy to respond to unexpected conditions.

2.12 Intelligent Underwater Robotic Systems

The development of intelligent underwater robotic systems represents the convergence of the technologies discussed above.

An intelligent inspection robot requires more than autonomous movement. It should be able to perceive its environment, estimate its position, identify relevant structures, recognise anomalies, plan actions and respond to changing conditions.

A possible intelligent inspection architecture therefore consists of:

- **Sensors:** Camera + Sonar + IMU + DVL + Depth Sensor
- **Sensor Fusion:** Vehicle State Estimation
- **Navigation:** Localization + Mapping + Path Planning
- **Perception:** Structure Detection + Object Recognition
- **Inspection:** Crack / Corrosion / Damage Detection
- **Decision Making:** Continue / Reposition / Inspect Again / Intervene

- **Manipulation:** Robotic Arm + End-Effector
- **Mission Reporting:** Defect Location + Condition + Inspection Data

This architecture reflects the direction identified by recent literature. Current research increasingly treats autonomous underwater inspection as an integrated perception and action problem rather than an isolated sensing problem. The 2026 systematic review of optical inspection of underwater critical infrastructure, for example, considers robotic platforms, image enhancement, datasets, detection, segmentation, quantitative assessment and 3D reconstruction as components of the wider inspection pipeline. ([ScienceDirect](#))

2.13 Conceptual Underwater Robotic Inspection System

Based on the literature reviewed, a conceptual underwater inspection system can be developed around a modular AUV or inspection class ROV. The proposed system could consist of five primary subsystems:

2.13.1 Mobility subsystem

The vehicle would use multiple thrusters to provide controlled movement in surge, sway, heave, pitch, roll and yaw.

2.13.2 Navigation subsystem

An IMU, DVL and depth sensor could provide the basic state-estimation inputs, supplemented by acoustic positioning or visual/sonar SLAM when appropriate.

2.13.3 Inspection subsystem

A forward facing or downward-facing camera could provide visual inspection data, while imaging sonar could support operation in low-visibility conditions.

2.13.4 AI perception subsystem

Collected images could undergo enhancement before being passed into a defect-detection or segmentation model. Potential outputs could include:

- Crack location
- Corrosion region
- Surface damage
- Marine growth
- Structural deformation
- Missing components

2.13.5 Intervention subsystem

Where the vehicle is designed for intervention, a manipulator could be added for selected tasks such as valve operation, object retrieval, cleaning or tool placement.

The conceptual system should not attempt to solve every underwater maintenance task. A more realistic research approach would be to select a limited inspection scenario and demonstrate the interaction between navigation, perception and inspection.

Example: **AUV approaches subsea pipeline → Navigation system maintains position → Camera/sonar detects pipeline → Vehicle follows pipeline → Camera captures inspection imagery → AI model identifies potential anomaly → Anomaly location is recorded → Robot generates inspection report**

2.14 Summary

The literature review indicates that underwater industrial inspection is transitioning from predominantly human operated and teleoperated approaches toward increasingly autonomous robotic systems. ROVs remain important because of their ability to support continuous communication and heavy intervention equipment, while AUVs provide greater mobility and autonomy for surveying and inspection. The emerging I-AUV concept attempts to combine these capabilities by giving autonomous vehicles the ability to physically interact with subsea infrastructure.

Sensing and navigation form the foundation of these systems. Cameras provide detailed visual information, sonar provides complementary perception under poor visibility, while IMUs, DVLs, acoustic positioning and SLAM techniques contribute to localization. Artificial intelligence and computer vision are increasingly being applied to image enhancement, anomaly detection, segmentation and condition assessment.

The literature also demonstrates that inspection and intervention represent different levels of robotic capability. Manipulation requires accurate target localization, vehicle stabilization, force interaction and motion planning. Communication and energy limitations further complicate autonomous operation, particularly for untethered AUVs.

Consequently, the development of an intelligent underwater inspection system requires the integration of **mobility, sensing, navigation, perception, artificial intelligence, communication, energy management and manipulation** rather than treating each technology independently. This provides the technical foundation for developing or simulating a conceptual robotic system capable of performing selected underwater inspection and monitoring tasks.

Features

1. **Nauert & Kampmann (2023) — Inspection and maintenance of industrial infrastructure with autonomous underwater robots**
Useful for your overall ROV/AUV industrial inspection argument. ([Frontiers](#))
[Read the paper](#)
2. **Systematic review of UUV-based structural health monitoring (2024)**
Particularly useful for your first objective and research-gap section. ([ScienceDirect](#))
[Read the review](#)
3. **Zhang et al. — Autonomous Underwater Vehicle Navigation: A Review (2023)**
Very useful for your navigation objective. ([ScienceDirect](#))
[Read the paper](#)
4. **Zhang et al. — Visual SLAM for Underwater Vehicles: A Survey (2022)**
Useful for visual navigation, SLAM and perception. ([ScienceDirect](#))
[Read the paper](#)
5. **Sivčev et al. — Underwater Manipulators: A Review (2018)**
This is one of the important papers for your manipulation/intervention objective. ([ScienceDirect](#))
[Read the paper](#)
6. **Ridao et al. — Intervention AUVs: The Next Challenge**
Very relevant to the transition from ROV intervention to autonomous intervention. ([ScienceDirect](#))
[Read the paper](#)
7. **SEAVENTION — Autonomous Subsea Intervention (2022)**
Particularly valuable for your objective concerning integration of navigation, perception, inspection and intervention. ([ScienceDirect](#))
[Read the paper](#)
8. **State of the Art in Key Technologies for AUVs (2025)**
Excellent recent source because it brings together **power, planning/control, perception, navigation, communication and AI** in one review. ([ScienceDirect](#))
[Read the paper](#)
9. **Tracking and Inspection of Subsea Pipelines and Cables Using Underwater Vehicles (2026)**
This is particularly relevant to your industrial-inspection focus because it compares **vision, sonar, electromagnetic/magnetic sensing and sensor fusion**. ([ScienceDirect](#))
[Read the paper](#)
10. **Optical Sensor-Enabled Automated Visual Inspection of Underwater Critical Infrastructure (2026)**
Very relevant to your AI/computer-vision objective because it covers the complete pipeline from **robotic acquisition → image enhancement → detection → segmentation → defect quantification → 3D reconstruction**. ([ScienceDirect](#))
[Read the paper](#)

CHAPTER THREE

METHODOLOGY